Barcelona Institute of Science and Technology

Universitat Pompeu Fabra Barcelona

Master Thesis

Master of Multidisciplinary Research in Experimental Sciences

Fractal dynamics and cancer growth

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Research Project

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Summary:

While many studies on cancer research have been published from a biological perspective, much less is known from a quantitative approach. Nowadays, applied mathematics is helping to fight cancer by integrating quantitative models of tumor growth, as the Gompertzian phenomenon and the power law, which behaviour is thought to have a fractal nature. In this project, developed at Institut de Física d'Altes Energies, it is considered a chaotic dynamical simulation model of tumor growth and metastasis process, sensitive to initial conditions, which is compared to the experimental breast cancer growth of two different cell tumor lines, namely MCF7 and MDA-231, obtained at Institut de Recerca Biomèdica Barcelona Foundation. Moreover, a three-dimensional strange attractor based on the designed model is constructed, which is dependent on the ratio between cell division and cell death probabilities, the tumor oxygen consumption and the releasement of pro-angiogenic factors, and the experimental parameters of which could be used to intercept the geometry of the tumor. A detailed analysis of the intercorrelation of the different variables is also provided. Hence, this project offers a new cutting-edge technique from a physical perspective using the properties of the fractal tumor's nature, the principle of minimum energy of which here is considered for the first time and proved successfully.

Keywords:

chaotic dynamics, principle of minimum energy, tumor evolution, self-metastasis

Presented on 14th of July, 2019
Registered at Universitat Pompeu Fabra

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Research Project

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I. INTRODUCTION

Cancer is one of the main causes of mortality in the world. Its conventional diagnostic is clinically detectable when it has reached a volume of 1 cm^3, which implies broadly 30 cell doublings from the progenitory cancer cell and a population of about 1 billion cells [1]. Considering volume doubling times of about 100 days, for example, the scenario is consistent with a preclinical time of 8 years. At the next 10 volume doublings (3 years), up to 10^3 cm^3, the cancer is reported to be lethal and passes through several different stages before it becomes large enough to cause death in patients [1–3]. Before reaching this last critical stage, large sums of money are invested into cancer research in order to find cures and improve existing treatments [3], although being focused on the erradication of all tumor cells to escape from the rapid regrowth that can be developed after chemotherapy is applied [4].

However, little is known about the nature of cancer and its quantitative approach. In comparison to the biological perspective, mathematics and physics have contributed relatively little to the area, whereas possess the hability to solve its epistemology [3].

Current tumor growth models are based on exponential kinetics, the most common of which are the Gompertzian phenomenon [4, 5] and the power law, which can both fit clinical data for breast and lung cancers, respectively [6]. Moreover, these models could be consequence of the fractal nature emerged by the natural result out of a necessity. Examples of this behaviour in nature can be found in the pattern formed by the branches of a tree, searching for an optimal configuration to expose the leaves to the sunlight; or in river basin geometries to minimize the flow transport's power expenditure [7, 8]. In those systems which nature is fractal, then, can be applied the principle of minimum energy, a fundamental principle developed to deal with the dynamics and the properties of complex systems like cancer, which fractal nature could emerge due to its constant nutrients supply necessity [9].

Furthermore and, by definition, the fractal nature is chaotic. Its dynamics, then, is sensitive dependent on the initial conditions, meaning that a small change in a deterministic variable of the system can result in large differences. While attractors in general dynamical systems are a set of numerical values in which the system will tend for specific initial conditions, being a fixed point or a periodic cycle, in chaotic dynamics are called *strange attractors*, which have scale-invariant structure and acquire non-integer Hausdorff dimensions.

Finally, in this introduction it is wanted to emphasyze the use of the chaotic dynamics as a trial to understand the epistemology of cancer. The enormous heterogeneity shown not only between but also in a same tumor line, could be the result of a set of different initial conditions in each case, leading the system to very different outcomes and showing a strange attractor dependent on experimental parameters that could be used to intercept its uncontrolled growth. This idea is reinforced by Larry Norton, who published the Gompertzian model for human breast cancer growth and suggests that

the basic phenomenon of cancer does not consist in genetic mechanisms but in the very geometry of the tumor [4]; and also reinforced by the *universal growth curve*, which was firstly published by G. West, J. Brown and B. Enquist for ontogenetic growths, plotting a dimensionless mass ratio versus a dimensionless time variable for a wide variety of determinate and indeterminate species in the same curve [10], and later on by C. Guiot, P. Degiorgis, P. Delsanto et al., who applied successfully the same model in different tumors [1].

It is reflected in this project the design of a chaotic dynamical simulation model of tumor growth and metastasis process, sensitive to initial conditions, which is compared to the experimental breast cancer growth of two different cell tumor lines, namely MCF7 and MDA-231. Moreover, it is considered for the first time and proved successfully the application of the principle of minimum energy in tumors, consequence of their fractal nature. The Hamiltonian used to verify this is able to model interactions between cells through volume and contact adhesion constraints, and the posterior Monte Carlo simulation of which results to minimize it. And finally, it is constructed a three-dimensional strange attractor based on the designed model, dependent on the ratio between cell division and cell death probabilities, the tumor oxygen consumption and the releasement of pro-angiogenic factors, the experimental parameters of which could be used to intercept the geometry of the tumor.

II. COMPUTATIONAL METHODS

Cancer as a complex system is composed by its entities, cells, that interact with themselves in a non-linear way. In this study, each cell will have two individual variables, namely cell division and cell death probabilities, and a global variable of the system, the metastasis process, consequence of the triggered mechanical response of cells in order to find an unoccupied space in the discrete lattice cube.

The tumor growth is simulated in a discrete cube with boundaries fixed at a determined number of nodes, L. Then, it evolves through its different stages: Initially, the tumor suffers a microavascular and exponential growth, such that cells proliferate as consume oxygen from pre-existent blood vessels. When oxygen concentration levels fall below a threshold, cells become hypoxic and start secreting pro-angiogenic factors. Endothelial cells from pre-existing blood vessels respond to these factors by forming new ramifications which geometry results to be fractal [2, 11]. Hence, an angiogenic switch is activated during the progression of the tumor to supply oxygen and nutrients when is required.

In this chapter, thus, are detailed the variables for simulating numerically the cancer growth with the goal, as said in the Introduction, to undertsand the optimal relation between the different variables describing the real cancer growth. For doing that, it is firstly described both cell division and cell death probabilities; then are explained the oxygen consumption and the releasement of pro-angiogenic factors through 2D diffusion processes, the numerical solution of which is provided on the discrete

domain of the lattice; then the metastasis process applied as a purely mechanical response, as mentioned above; and finally is detailed the Hamiltonian used to calculate the total energy of the tumor, based on the Glazier-Graner-Hogeweg (GGH) model, with which can be proved the principle of minimum energy through a Monte Carlo simulation. This chapter allows, thus, to perform simulations that would be then contrasted with experimental data.

For comodity, time derivatives of variables x are written as $\dot{x}$, and the vector $\vec{x}$ indicates three dimensional space dependence, as well as t indicates time dependence.

A. Cell division and cell death probabilities

At each simulation time step all cells will have individual probabilities for both cell division and cell death. Moreover, in both cases are neglected all but their main contributions: the ratio of the relative consumption of oxygen that can be taken at the specific coordinates of the cell compared to the maximum can consume the tumor before becoming hypoxic (Eq. 1). The higher the ratio, the more hypoxic is becoming the tumor. Details of the consumption of oxygen are explained in II C.

$$C = \frac{c(\vec{x}, t)}{c_{max}} \tag{1}$$

Thus, cell death probability (d), Eq. 2, is dependent on the ratio of oxygen consumption C and a resistance factor, α, an intrinsec parameter of the tumor that adjusts the maximum death probability of cells. The greater C, the more approximated cell death probability to α.

$$\dot{d} = \alpha C \tag{2}$$

It is considered in simulations two types of cells, namely condensing and non-condensing, randomly distributed along the tumor, and the use of which are applied in proliferation of cells and their interactions to calculate the total energy of the tumor, II F. Thus, the cell division probability (b), Eq. 3, takes the opposite contribution to the consumption ratio, $1 - C$, but also an additive contribution, the condensing factor γ, that depends on the type of cell: positive if condensing and negative otherwise. Moreover, and analogously to d, cell division has also a dependence on an intrinsec parameter, the growth factor β, that adjusts the maximum division probability of cells. In this case, the less C, the more approximated cell division probability to β.

$$\dot{b} = \beta\Big(1 + \gamma - C\Big) \tag{3}$$

All cells, then, have determined cell division and cell death probabilities. At the beginning all cells have great b and low d, but when the tumor becomes hypoxic the nucleus have greater d, whereas

in the periphery b remains significant. If the cell dies, its occupied site is available again, whereas if the cell has been duplicated, this will equiprobably occupy one of its 1st and 2nd order nearest neighbors (it is used up to second order to reduce artifacts due to the anisotropy of the square lattice [12]). As tissue dimensions can be calculated through fractals, the number of nearest neighbors involved requires to be taken carefully. When allowed to growth, a normal tissue (which has a low fractal dimension of 2.5) drops quickly, whereas in the case of cancer tissue (which has a high fractal dimension of 2.98) maintains a high density, thus feeding an even faster growth [5]. If a length L is increased in a three-dimensional object, its volume will be increased as L^3. However, the cellular mass increases as L^λ, being λ its fractal dimension. In the case of tumors, thus, the ratio between cellular mass and volume is kept approximated to the unit. Hence, simulations can take 1st and 2nd order nearest neighbors from the cubic lattice, a total of 18.

After the angiogenic switch has been turned on, the choose of nearest neighbors won't be equiprobable anymore. Then, pro-angiogenic factors have been released from the surface, detailed in II B, and the free available oxygen in the tumor has been positively modified. The duplicate cell in this new case will choose the nearest neighbor which most free available oxygen concentration has (or less consumption of it). This is not only a way to approximate the place where the fractal ramifications emerged from angiogenesis are likely to grow, but also to promote chemotaxis in cells, the movement of which is in response to the chemokines released.

B. Pro-angiogenic factors: chemokines and chemotaxis

The releasment of pro-angiogenic factors (ϕ) is carried out after the angiogenic switch has been turned on. These factors include chemokines, signaling proteins secreted that promote chemotaxis, the movement of cells through which follows an oxygen concentration gradient. While in the avascular growth the evolution of the tumor was randomly developed through its 18 possible nearest neighbors, in angiogenesis it is not random anymore.

In the simulations is carried out this angiogenic switch by secreting in a certain thickness of the approximated spherical shell of the tumor a diffusion process (Eq. 4), the diffusion coefficient d_{ch} of which determines how fast this is performed. Moreover, the ratio between the number of current cells in the tumor (N) and the number of cells at which hypoxia is present in the tumor and angiogenesis can start developing (N_a) is added to limit this diffusion process until required.

$$\dot{\phi} = \frac{N}{N_a} d_{ch} \nabla^2 \phi(\vec{x}, t) \tag{4}$$

Below is detailed the effect of these diffusion process on the oxygen consumption.

C. Oxygen consumption and hypoxia

Cells consume oxygen to carry out their glucose uptake through the so-called Warburg effect. Whereas normal cells produce their energy through mitochondrial oxidative phosphorylation, most cancer cells produce their own energy through a high glycolysis rate followed by a lactation fermentation process, even when there is plenty of oxygen in the nearby environment [2, 13].

Initially, cells will have free oxygen available, the consumption of which will be minimum, c_{min}. When growing enough, this consumption will overcome a threshold in the nucleus, c_{max}, such that won't be able to maintain cells in the zone and becoming, thus, hypoxic. This oxygen consumption (c) is simulated, analogously to ϕ, through a diffusion process, Eq. 5. Moreover, when pro-angiogenic factors are released, these are substracted from c with an effectiveness factor (δ) that indicates which fraction of free oxygen is available again.

$$\dot{c} = d_{ox}\nabla^2 c(\vec{x}, t) - \delta\phi(\vec{x}, t) \tag{5}$$

In the results section it is analyzed the interaction between both diffusion processes. But as the diffusion equation is solved in the continuous domain, below is developed the numerical solution to be applied on a discrete lattice.

D. Numerical solution of diffusion processes in a discrete domain

For comodity and to observe the evolution of both c and ϕ diffusion processes, these are approximated in 2D. Thus, the two-dimensional diffusion equation (Eq. 6), which is derived from the continuity equation, is applied in continuous fields.

$$\frac{\partial U}{\partial t} = D\nabla^2 U = D\Big(\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}\Big) \tag{6}$$

A simple numerical solution of the domain of the cube $0 \leq x < L$, $0 \leq y < L$ approximates the $U(\vec{x}, t)$ by the discrete function $u^n_{i,j}$ where $x = i\Delta x$, $y = j\Delta y$ and $t = n\Delta t$. Applying finite difference approximations it is reached a state of the system at time step $n + 1$ (Eq. 7), which variable is taken into account for both c and ϕ diffusions in addition to their diffusion coefficients.

$$u^{n+1}_{i,j} = u^n_{i,j} + D\Delta t\Big[\frac{u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}}{(\Delta x)^2} + \frac{u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1}}{(\Delta y)^2}\Big] \tag{7}$$

In results section it is applied this numerical solution to both diffusion processes, with a different time step in each, n_{ox} and n_{ch}, assuming a priori that times at which consumption is taken and pro-angiogenic factors are released are not the same.

E. Metastasis process

In this project it is wanted to emphasize the idea of self-metastatic tumor, proposed by Larry Norton [5], a different perspective in modelling terms of cell invasion and metastasis carried out due to its large size [2]. Thus, metastasis will be understood as a purely mechanical response that is applied since the tumor starts developing.

In simulations, the metastasis process is triggered when the chosen nearest neighbor of the duplicate cell is already occupied. Then, is required to upload nearest neighbors each time until an unoccupied site is found. However, when reached, if the last occupied site has only a single 1st order nearest neighbor, that is to say, only one over its six available, then the duplicate cell is detached from the tumor, thus, becoming a metastatic cell.

By applying this method the metastatic cell fits with two requirements it has, namely the lose of cell-to-cell adhesion (resorting the tumor until finding its site) and its detachment due to the low adhesion that presents the environment (the condition to be detached of the single 1st order nearest neighbor applied to the last candidate), usually shown on the surface [2].

In the results section it is analyzed the production of metastatic cells depending on the conditions applied in the tumor.

F. The principle of minimum energy in tumors

In order to prove the principle of minimum energy in tumors, consequence of their fractal nature, it is required to model a Hamiltonian able to implement properties, behaviours and interactions via constraints in the tumor.

M. Swat, G. Thomas, J. Belmonte et al. applies the Glazier-Graner-Hogeweg (GGH) model, which canonical formulation is derived from statistical physics, to construct a Hamiltonian, Eq. 8, that represents the effective energy of the tumor [12]. This energy does not represent the physical energy of cells, but is a way to produce a desired set of cell behaviours with only defined volumes area and interactions via contact adhesion (through 2nd order nearest neighbours, see II A).

$$H = \sum_{\substack{\vec{i},\vec{j} \\ neighbors}} J\Big(\gamma(\sigma_{\vec{i}}), \gamma(\sigma_{\vec{j}})\Big)\Big(1 - \delta(\sigma_{\vec{i}}, \sigma_{\vec{j}})\Big) + \sum_{\sigma} \Big[\lambda_{vol}\Big(v(\sigma) - V_t\Big)^2\Big] \tag{8}$$

The first sum of the equation is applied over all pairs of nearest neighbors cells of lattice sites $\vec{i}$, $\vec{j}$, and represents the contact energy between nearest neighbors cells to implement adhesive interactions. The parameter J, thus, is the boundary energy per unit contact area for a pair of cells, represented in the equation as σ.

In simulations, then, three types of J are used, depending on the pair of type of cells that interacts, distinguished by their condensing factor γ (see II A), namely condensing and condensing interaction (J_{cc}), non-condensing and non-condensing interaction (J_{nn}), and condensing and non-condensing interaction (J_{cn}). Thus, higher contact energy between cells will represent greater repulsion between them, whereas lower contact energy will represent greater adhesion. The interactions simulated will always satisfy $J_{cc} < J_{cn} < J_{nn}$.

The second sum represents the effective energy due to the volume constraint. In simulations, all cells will have associated a random volume area, v, the value of which is up to its target volume area, V_t. Deviations of v from V_t will increase the effective energy, thus penalizing these deviations. The parameter λ_{vol} acts as a Young modulus and is taken fixed in simulations.

As the type of cell and its associated volume have been distributed randomly along the tumor, the principle of minimum energy can be proved by obtaining a final configuration from the initial obtained once the tumor is developed, carrying out a Monte Carlo simulation. This is simply executed by applying the Metropolis algorithm, Eq. 9. Before each trial, it is selected a random cell and inverted its type. Then, the probability to accept the change in its type is calculated as shown, by using the difference between new and old configurations of the tumor energy, calculated with Eq. 8, and by using a control parameter, T, fixed in simulations carried.

$$P(\sigma_{\vec{i}} \rightarrow \sigma_{\vec{i}'}) = \begin{cases} 1, & \text{if } \Delta H \leq 0 \\ e^{-\frac{\Delta H}{T}}, & \text{if } \Delta H > 0 \end{cases} \tag{9}$$

The greater $\Delta H/T$, the more rigid and non-mobile cell with their nearest neighbors. This can be accomplished with greater number of J_{cc} and low deviations of v from V_t. On the contrary, the lower $\Delta H/T$ will mean absence of a constraint enough to maintain the integrity of the borders between the cell and its nearest neighbors. This is accomplished, then, with a greater number of J_{nn} and high deviations of v from V_t.

In results section is analyzed the variation of energy over the steps applied in the Monte Carlo simulation, as well as the evolution of cell types through it.

III. EXPERIMENTAL METHODS

Complementary to simulations, it has been determined experimentally the evolution of two different breast tumor cell lines, namely MCF7 (known to be hormone-dependent) and MDA-231 (hormone-independent) [14, 15], in six BALB/c nude mice each group in ad libitum conditions. In MCF7 group was injected one mammary fat pad per mouse, whereas in MDA-231 group two mammary fat pads were injected.

In this chapter, thus, are provided real parameters of experimental growth that will help to adjust parameters from simulations. Below is detailed the cell cultured preparation, the volume measure applied and the study of tumor proliferation.

A. Cell culture

Both MCF7 and MDA-231 tumor cell lines were thawned and cultured under standard conditions (that is to say, 37℃, 5% of carbon dioxide and humidified atmosphere). DMEM medium was supplemented with 10% of fetal bovine serum (FBS), penicillin/streptomycin and glutamine.

The medium is recommended to be changed after 3 days, when its color has changed from red to orange, indicating that medium is acidified. Then, a new passage is carried by absorbing to vaccum the medium, leaving the tumor cells deposited at the plate's base, and renewing the mix medium with 2 mL of trypsin to let cells be in suspension again. After the passage, the proliferation of tumor cells can be observed qualitatively through an optical microscope.

Cells were quantized through Neubauer chambers with 10 mL of micro isotonic and 10 mL of tumor cells. It was averaged the number of cells per mL that were counted in each grid and extrapolated to 30 mL that were prepared (26 mL of mix and 4 mL of trypsin). In each tumor cell line, MCF7 and MDA-231, were obtained, respectively, 10.23M and 22.86M cells. As were injected 1M cells per mice, for MCF7 group were used all 30 mL, whereas for MDA-231 were used 26.2 mL (this implies 20M tumor cells available to be injected).

After quantifying, were centrifuged both mixes to obtain the tumor pellet. With it, was added 250 μL of PBS and 250 μL of Matrigel (which supports the tumor growth) for the MCF7 case, whereas it was doubled both quantities for the MDA-231 case, due to its doubled number of cells. Then, both tumor compounds were ready to be injected.

B. Volume measurement and tumor proliferation

Measurements of tumor length (L) and width (W) dimensions were obtained weekly with a digital caliper to calculate the tumor volume, Eq. 10.

$$V_{tumor} = \frac{\pi}{6} L W^2 \tag{10}$$

After sacrifices were carried, could be studied the tumor proliferation by performing an immunohistochemical test to detect Ki-67, a proliferation marker that is only detected in the cell nucleus [14]. The quantification of proliferation percentage in tumors was obtained through the free software QuPath. The analysis was performed for each tumor in three regions, namely the periphery, the nucleus and

the whole surface of the section. This analysis was performed to observe if there's any significant change in the distribution of the tumor proliferation.

In results section is analyzed the tumor volume evolution and proliferation, as well as it is applied its pertinent statistics.

IV. RESULTS

Firstly, simulations performed have fixed variables, namely $c_{min} = 100, c_{max} = 500, \gamma = 0.1, N_a = 500, d_{ox} = 2, d_{ch} = 1000, \delta = 1.2, n_{ox} = 100$ and $n_{ch} = 50$. With them, in figure 1 it is represented the oxygen evolution in the tumor until hypoxia is reached, as well as the effect that produces the releasment of pro-angiogenic factors in the tumor.

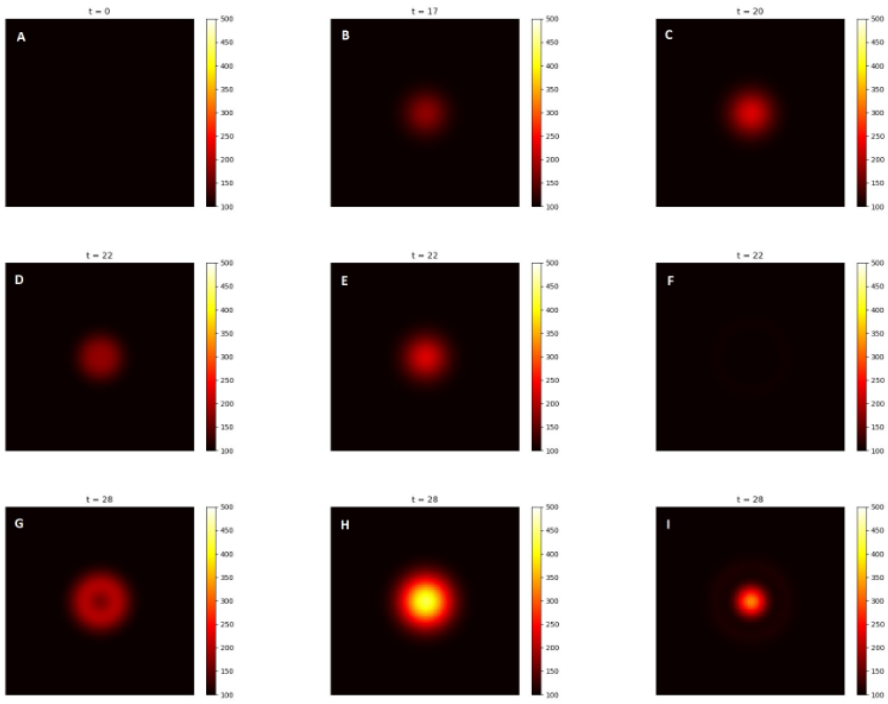

FIG. 1: **A, B, C:** Evolution of oxygen consumption (c) over time. While in A tumor starts developing and c is minimum, in B starts being visible after 17 simulation times, and in C the tumor has reached almost hypoxia, with higher c. **D, E, F:** E represents the tumor when has reached hypoxia and the angiogenic switch has been turned on. Thus, D are the pro-angiogenic factors released (ϕ) on the 30% thickness of the tumor surface shell. F represents the new c after ϕ has intervened. In this case, as the tumor started angiogenesis at a low diameter, ϕ is able to penetrate to the very centre of the tumor, refilling it with oxygen supply. **G, H, I:** Analogously to the previous case, H represents an hypoxic tumor with high diameter (greater β and lower α). G represents ϕ, and I its effect on H. Now, due to the large tumor, ϕ will penetrate until certain depth, helping the tumor to provide oxygen supply but unable to refill its center, which will become necrotic over time.

The resistance (α) and growth (β) factors were modified to observe different behaviours and heterogeneity in tumors due to the stochastic nature of simulations. Thus, in figure 2 can be compared

both simulation and experimental heterogeneities in tumors. On the left, tumor population of simulations is shown. For low α (0.3), it is observed that there's an optimal β (approx. 0.7) to let tumor growth. Thus, and counterintuitively, a high value of β (0.9) implies a tumor diameter large enough when the angiogenic switch is turned on that is not able to be refilled again with oxygen supply through ϕ. On the right, experimental breast tumor growths of MCF7 and MDA-231 is shown. Predominantly, MCF7 tumors are less aggresive than MDA-231. The ratio of growth between groups were statistically significant at $p = 0.03$. In fact, bigger MCF7 tumors had the equivalent size of smaller MDA-231.

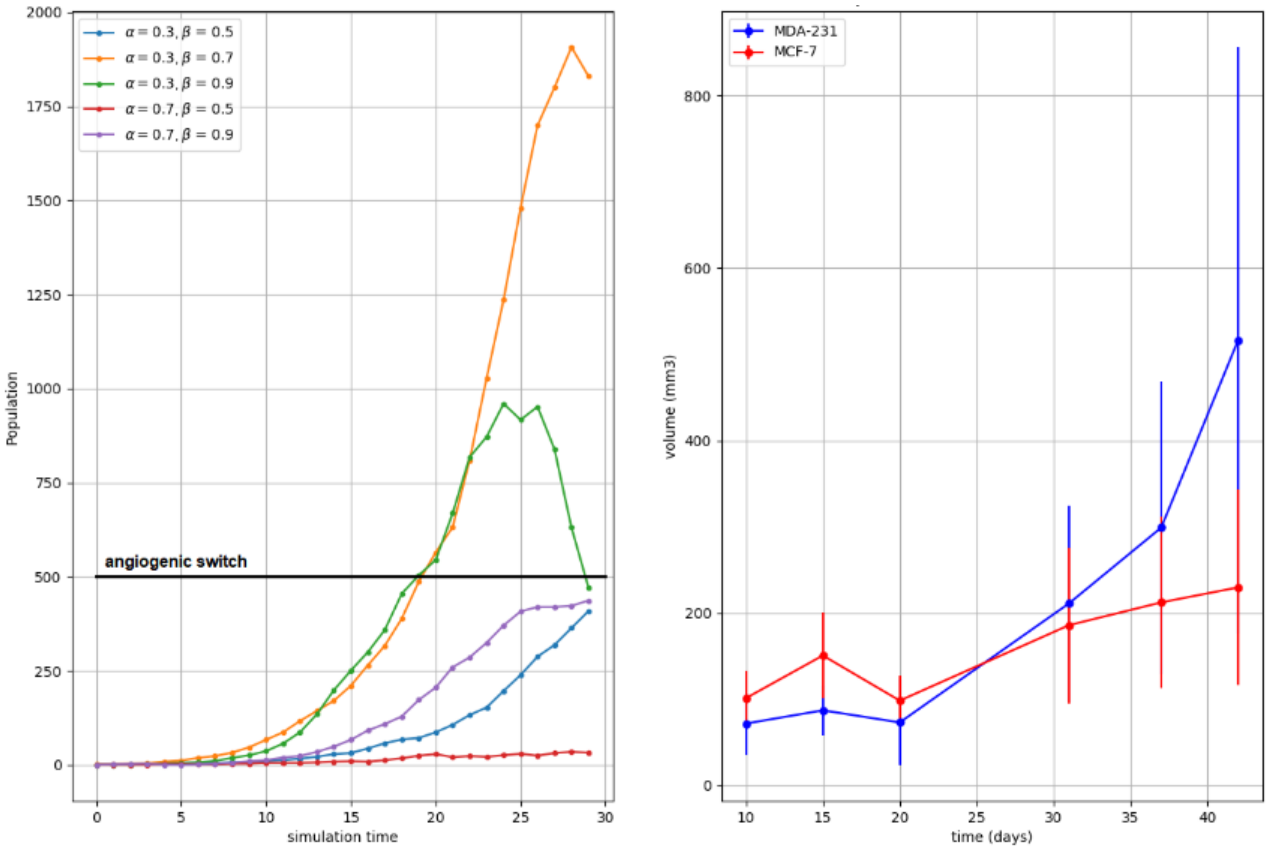

FIG. 2: **Left:** Population of simulated tumor evolution over time for different values of α and β. The angiogenic switch is stablished at $N_a = 500$. **Right:** Experimental breast tumor evolution for both MCF7 (red) and MDA-231 (blue) cell tumor lines, the standard deviation of which shows a significant heterogeneity in each. Sample sizes for MCF7 and MDA-231 groups were $N = 6$ and $N = 10$, respectively.

Cell division probability (b) simulated was compared to the experimental tumor proliferation, both of which are represented in figure 3. In A is shown simulated b for three different pairs of (α, β) from figure 2. Usually b starts at its maximum value and decays when C is approx. the unit. Depending on how fast hypoxia is reached, the decay of b can be abrupted, as it was the case of (0.3, 0.9). From C and D images it could be inferred the distribution of proliferation in tumors. Whereas the global proliferation between groups was not statistically significant ($p = 0.47$), the differences between nucleus and periphery were significant in MDA-231 ($p = 0.01$) and not significant in MCF7 ($p = 0.37$).

Thus, and as shown in C and D images, MCF7 tumors have a homogeneous proliferative distribution, whereas MDA-231 result to have an heterogeneous distribution. This may indicate that MDA-231 tumors have low α (0.3) and high β (0.9), with nucleuses that, as simulations shown, are probably necrotic.

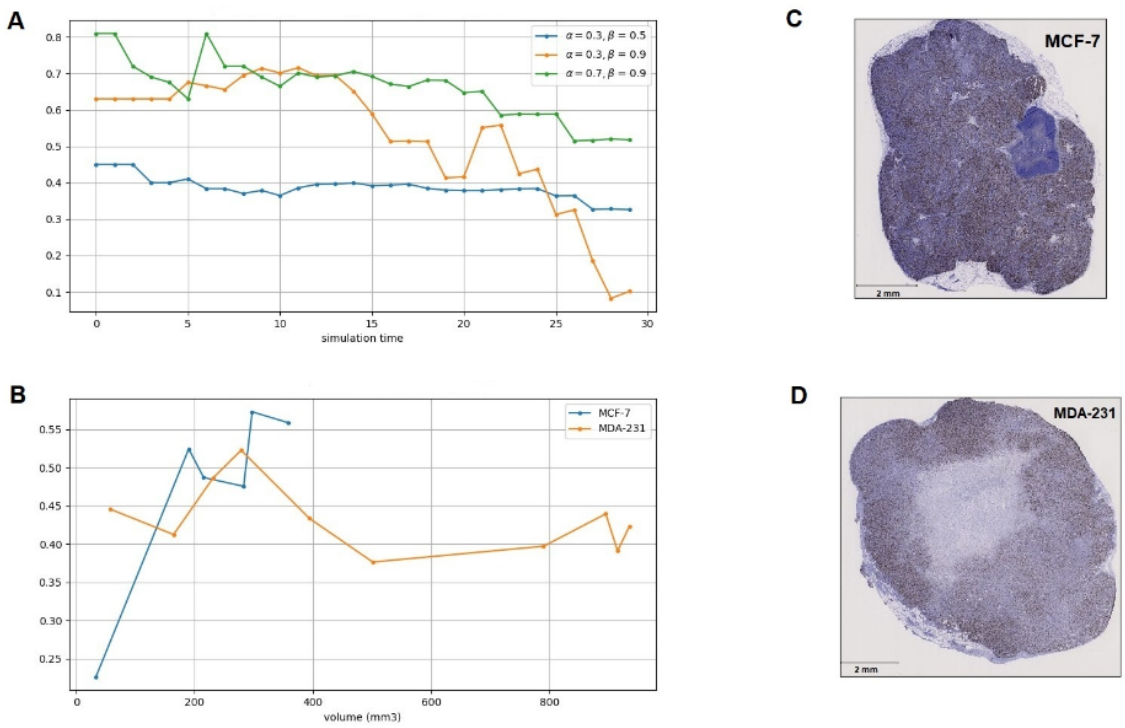

FIG. 3: **A:** Simulated cell division (b) over simulation time for different pairs of α, β. Each point represents the average b of current cells in tumor. **B:** Experimental proliferation from the whole surface of the section for MCF7 (blue) and MDA-231 (orange) cell tumor lines over their volumes in post-sacrificed time (mm^3). **C, D:** Representative images of MCF7 and MDA-231 tumor sections, respectively. Dark zones are proliferative, whereas light zones aren't.

Metastasis process is, as explained in II E, the global variable of the system and a mechanical response produced since the tumor starts developing, a concept denominated self-metastasis. As shown in figure 4, the metastatic cells are those produced in tumors simulated in figure 2. From it, can be inferred that those tumors with large size are those with greater production of metastatic cells, a behaviour that fits with experiments carried [2, 5]. In simulations, thus, it is reinforced the self-metastatic tumor: Cancer doesn't reach a size at which becomes metastatic, it is large because of its self-metastatic condition.

The principle of minimum energy, consequence of the fractal nature of the tumor, could be proved successfully by applying the Hamiltonian used in Eq. 8 by using defined volumes area and interactions via contact adhesion in cells, once the tumor is developed. Then, through a Monte Carlo simulation, the total energy of the tumor is minimized, as observed in figure 5 (left). In this simulation has been used a target volume area with length $L = 20\mu m$, which is the average size of cancer cells

[13], boundaries energy $J_{cc} = 2, J_{cn} = 11, J_{nn} = 16$ and $\lambda_{vol} = 2$. Moreover, in figure 5 (right) it is shown that, while total energy of the tumor is minimized, condensing clusters of cells are formed. This may indicate a phase transition in the tumor in order to stabilize it.

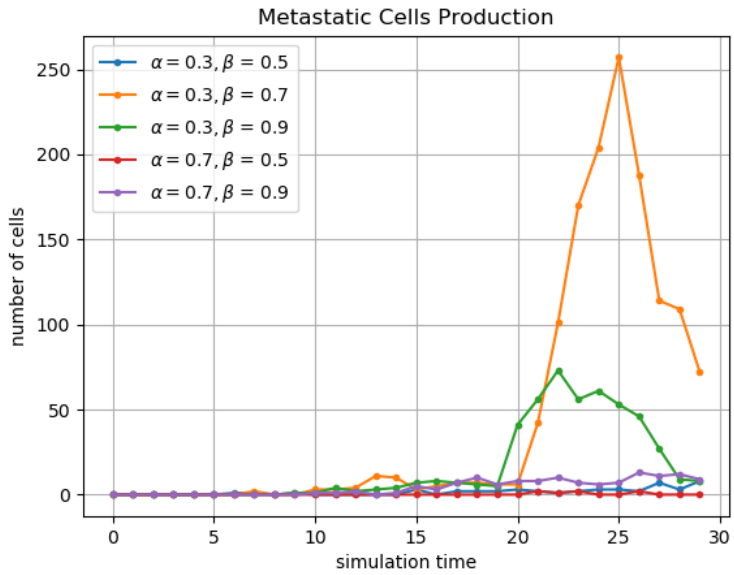

FIG. 4: Production of metastatic cells from simulated tumors of figure 2.

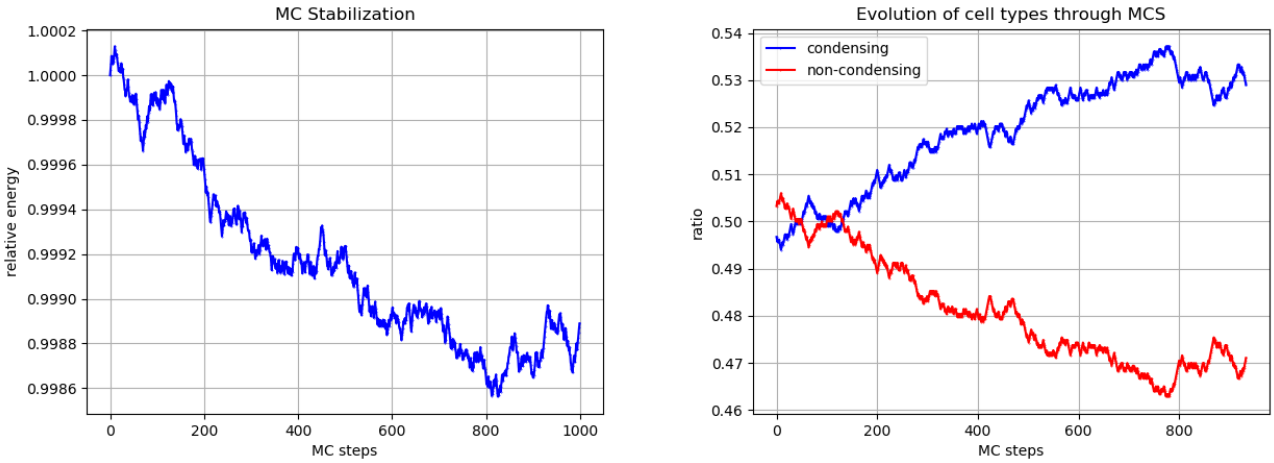

FIG. 5: **Left:** Minimization of the total energy of the tumor. **Right:** Evolution of cell types through the Monte Carlo simulation: condensing (blue) and non-condensing (red).

Finally, once proved the nature of cancer, it is wanted to see if a modified version of the system could lead to a chaotic description of tumor growth. In order to do it, perturbations were introduced in the model, Eq.11, namely $\epsilon, \eta, \omega < 1$. This set of coupled equations lets construct a strange attractor, figure 6, dependent on the ratio between cell division and cell death probabilities, R, oxygen consumption c and the pro-angiogenic factors released ϕ.

$$\begin{cases} \dot{R} = \frac{\beta\epsilon^2}{\alpha C}(1 + \eta\gamma - C) \\ \dot{c} = \eta(\delta\phi - c) \\ \dot{\phi} = -\omega R \end{cases} \tag{11}$$

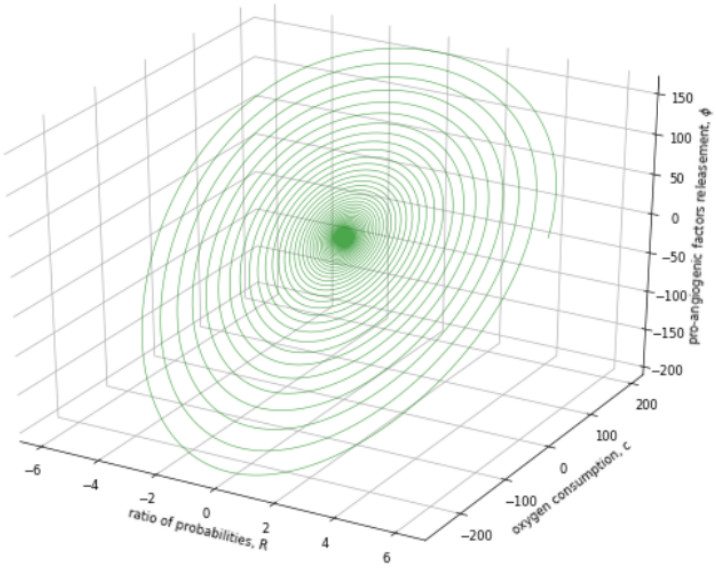

FIG. 6: 3D Tumor strange attractor dependent on R, c and ϕ.

Thanks to the confirmed fractal nature of cancer developed in this project through the principle of minimum energy and its consequent strange attractor shown in figure 6, experimental parameters that can modify R, c or ϕ and discrupt, thus, the attractor could also, and this is yet object of study (see Discussion section) , disrupt the geometry of the tumor and make it incapable to survive.

V. CONCLUSIONS

In this project it has been developed a chaotic dynamical simulation model of tumor growth, which depends on cell division (b) and cell death (d) probabilities, the oxygen consumption (c) and the releasment of pro-angiogenic factors (ϕ). Both probabilities have intrinsec factors, α and β, with which, together with the interaction between c and ϕ diffusion processes, is proved heterogeneity in tumors. The perturbations added together with the set of coupled equations let construct a 3D strange attractor, dependent on the ratio between cell division and cell death probabilities (R), c and ϕ.

Moreover, could be inferred experimental tumor evolution and proliferation to simulations carried. MDA-231 tumors have a heterogeneous proliferation distribution, greater in their periphery. This may indicate necrosis in their nucleuses and, thus, low α and high β parameters, which fit in simulations performed with 0.3 and 0.9 values, respectively.

It is reinforced the self-metastatic tumor concept, the metastasis process of which is a purely mechanical response of the tumor since starts developing and with a greater production at a larger size of it, which behaviour matches with experimental data found in literature [2, 5].

Finally, the application of the principle of minimum energy in tumors, consequence of their fractal nature, has been considered for the first time and proved succesfully by taking a Hamiltonian which constrains depend only on the volume of cells and their contact adhesion interaction.

VI. DISCUSSION

Firstly, here it is remarked the importance of having proved the principle of minimum energy in tumors. This is the first indicative of its fractal nature and, from all the properties that it entails, it is constructed a 3D strange attractor that can be used to disrupt the geometry of the tumor and make it, then, incapable to survive. Thus, further work is required by using *control of chaos* techniques and suppress or weak the attractor. One example is by using OGY-methods [3], used to intercept the attractor and, thus, the tumor evolution.

While simulations have the powerful tool to recreate complex systems, in this project parameters used were enough to prove the aims proposed, but insufficient to model a real tumor, mainly due to computational costs. The angiogenic switch is known to be turned on at its 6.6 mm tumor diameter [16], far away from the 500 nodes used in simulations.

It is also important to remark that tumors saturate as their final stage. This can be accomplished when cell division and cell death probabilities match, the paired factors α and β of which will have an specific time to saturate in each case. Further work can be done here, by finding intrinsec parameters α and β for each tumor, as done here with MDA-231 cell tumor line, and time required to achieve each of its stages, namely angiogenesis and saturation.

Although is not object of study here, entropic effects need to be considered to let simulations performed be more consistent. In [13] it is achieved entropic effects in normal and cancer cells, and found that the external entropy density production is greater in cancer cells, which main contribution is due to the heat irreversible release. Taking this study into account and the Hamiltonian used here to prove the principle of minimum energy, Eq. 8, MaxEnt methods should be performed to reveal if the constrains applied in this project are enough to prove the fractal nature of cancer.

Moreover, the cellular automata is used as an evolutionary game applied also in cancer [17]. Simulations performed in this project can be compared with conditions stablished in *The Game of Life*, well-known to simulate fractal structures through the interactions of their entities. In special, here it is remarked the glider spaceship, the object's structure of which is rejected from the system and propagated through the space, just as metastasis cells do.

Acknowledgments

I would like to thank Pere for the opportunity to develop this challenging project, as well as Roger for letting me experience for the first time with cells and mice in the animal facilities. Thank you also to Fernando and Marc, from IRB, for their patience during my stay in the lab, to my Master colleagues for helping me with bio-questions I had during the project, and to my family for their support and blind faith on me.

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